



ISO/TC 12 "Quantities and units"
Secretariat: SIS
Committee manager: Radhe Sven Mr



PPT Introducing CalConnect (Ronald)

Document type	Related content	Document date	Expected action
Meeting / Liaison report	Meeting: VIRTUAL 22 May 2025	2025-05-22	INFO by 2025-05-22

CALCONNECT

Introducing CalConnect

ISO/TC 12

Ronald Tse, Vice President, CalConnect

May 22, 2025

CalConnect background: SDO for C&S / Contacts

- Incorporated January 2004
 - 501(C)6 Mutual Benefit Association
 - Non-Profit, member-driven
- Focused on
 - Calendaring & Scheduling
 - Contacts
 - Time zones
- Original specs from Versit Consortium, IP donated primarily to IETF in late 1990s
- CalConnect formed to improve C&S technologies and products
 - Standards and specifications
 - Interoperability testing
 - Collaboration between C&S professionals

CalConnect standards are in prevalent use today



- Liaisons with ISO, ITU, IETF, M3AAWG
- Reciprocal membership with OASIS
- Our iCalendar, vCard standards used on over 2.5 billion devices
- CalDAV, CardDAV powers the largest cloud services

All Calendar / vCard-related documents and standards since 2005 have been work products of CalConnect in one form or another

CalConnect works globally

- International Scope
 - North America
 - United Kingdom
 - Western / Eastern Europe
 - Asia and Far East
 - Pacific Rim / Australia
- Currently about 30 members
 - Vendors of all sizes
 - Research Universities
 - Companies & Government organizations
 - Individual members

CalConnect liaisons with ISO

- ISO/TC 20/SC 6
 - ISO/TC 37 +{SC 4, SC 5}
 - ISO/TC 46 +{SC 4, SC 9}
 - ISO/TC 154
 - ISO/TC 211
 - ISO/IEC JTC 1/SC 27
- ISO 24229:2022 Information and documentation — Codes for written language conversion systems
- ISO 690:2021 Information and documentation — Guidelines for bibliographic references and citations to information resources
- ISO 8601-1:2019 Date and time — Representations for information interchange — Part 1: Basic rules
- ISO 8601-2:2019 Date and time — Representations for information interchange — Part 2: Extensions
- ISO 8601-1:2019/Amd 1:2022 Date and time — Representations for information interchange — Part 1: Basic rules — Amendment 1
- ISO 8601-2:2019/Amd 1:2024 Date and time — Representations for information interchange — Part 2: Extensions — Amendment 1
- ISO 34000:2023 Date and time — Vocabulary
- ISO 19135:2024 Geographic information — Procedures for item registration

Comments on ISO 80000-2

- In 2023, CalConnect submitted 10 pages of comments to ISO 80000-2:2019
- Systematic review indicated 41 comments from DE and 1 from PL
- Potential to revise?

CC-01	Item 2-13.1	ed	The number "2,718 81 28" is not correctly formatted with the numeric grouping practice of 3 digits as described in ISO DIR 2. It should have been "2,718 812 8". 1.3 Exponential and superlative functions Complex exponents can be used, in particular for the base e. Table 3 – Symbols and notations for exponential and superlative functions <table> <tr> <th>Symbol</th> <th>Meaning</th> <th>Remarks and examples</th> </tr> <tr> <td>e^x</td> <td>Exponential function</td> <td>$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = 2.718\,281\,828\ldots$ This number is called Euler's number.</td> </tr> </table>	Symbol	Meaning	Remarks and examples	e^x	Exponential function	$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = 2.718\,281\,828\ldots$ This number is called Euler's number.	The correct numeric format is shown with demonstrable usage in Clause 4 "e = 2,718 281 828".
Symbol	Meaning	Remarks and examples								
e^x	Exponential function	$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = 2.718\,281\,828\ldots$ This number is called Euler's number.								
CC-02	Item 2-17.3	ed	Immediately after "sin", there should be a thin space as per ISO 80000-2, Clause 4 ("there shall be a thin space between the symbol for the function and the argument"). The thin space is verified to be missing in the PDF content tree. Table 4 – Symbols and notations for trigonometric functions <table> <tr> <th>Symbol</th> <th>Meaning</th> <th>Remarks and examples</th> </tr> <tr> <td>$\sin$</td> <td>Sine function</td> <td>$\sin x$ is the ratio of the opposite side to the hypotenuse in a right-angled triangle.</td> </tr> </table>	Symbol	Meaning	Remarks and examples	$\sin$	Sine function	$\sin x$ is the ratio of the opposite side to the hypotenuse in a right-angled triangle.	Add a thin space after the "sin" function.
Symbol	Meaning	Remarks and examples								
$\sin$	Sine function	$\sin x$ is the ratio of the opposite side to the hypotenuse in a right-angled triangle.								
CC-03	Item 2-20.7	ed	The "sin" is missing the "thin space" immediately after "sin" (the function name) as required by Clause 4. The thin space is verified to be missing in the PDF content tree. Table 5 – Symbols and notations for inverse trigonometric functions <table> <tr> <th>Symbol</th> <th>Meaning</th> <th>Remarks and examples</th> </tr> <tr> <td>$\arcsin$</td> <td>Inverse sine function</td> <td>$\arcsin x$ is the angle whose sine is x.</td> </tr> </table>	Symbol	Meaning	Remarks and examples	$\arcsin$	Inverse sine function	$\arcsin x$ is the angle whose sine is x .	Add a thin space after the "sin" function.
Symbol	Meaning	Remarks and examples								
$\arcsin$	Inverse sine function	$\arcsin x$ is the angle whose sine is x .								
CC-04	Item 2-6.4	ed	The last formula under Remarks has a full stop at the end of the formula. This is confusing because it is unclear whether the full stop belongs to the formula, as the formula is not an "inline formula", but starts off a new line after the colon (":").	This formula should either be placed inline, or the full stop moved to the next line (or just removed).						

Ongoing projects: Calendaring and scheduling

- TC CALENDAR
 - JSCalendar has been published in IETF RFC 8984
 - iCalendar format in JSON format
 - iCalendar Tasks
 - Richer model for task management
 - iTip revision
 - Handling iCalendar content inside modern emails
 - Calendar Dev Resources
 - Resources for iCalendar developers
- CC 51003 Series in iCalendar
- CC 51004 iCalendar Relationships
- CC 51005 Calendar Subscription Upgrades
- CC 51006 Consensus scheduling -- iCalendar vPoll component

Ongoing projects: Date and time

- TC DATETIME
 - CC 34100 “Reference time scales”
 - CC 34200 “Timezones”
 - CC 34300 “Calendar codes” (ISO/AWI 34300)
 - CC 18013 “Date and time -- Representation of value metadata”
 - Calendar, time and timezone codes
 - (ISO: with ISO/TC 154/WG 5)

Ongoing projects: Documents

- TC PUBLISH
 - CC 36100 Standardization document – Metamodel (also ISO/AWI 36100)
 - CC 36200 Standardization document – Metadata (also ISO 36200)
 - CC 36300 Standardization document – Representation in XML (also ISO 36300)
 - CC 36010 Lightweight document – Metamodel
 - (ISO: with ISO/TC 154/WG 4)

Ongoing projects: vCard

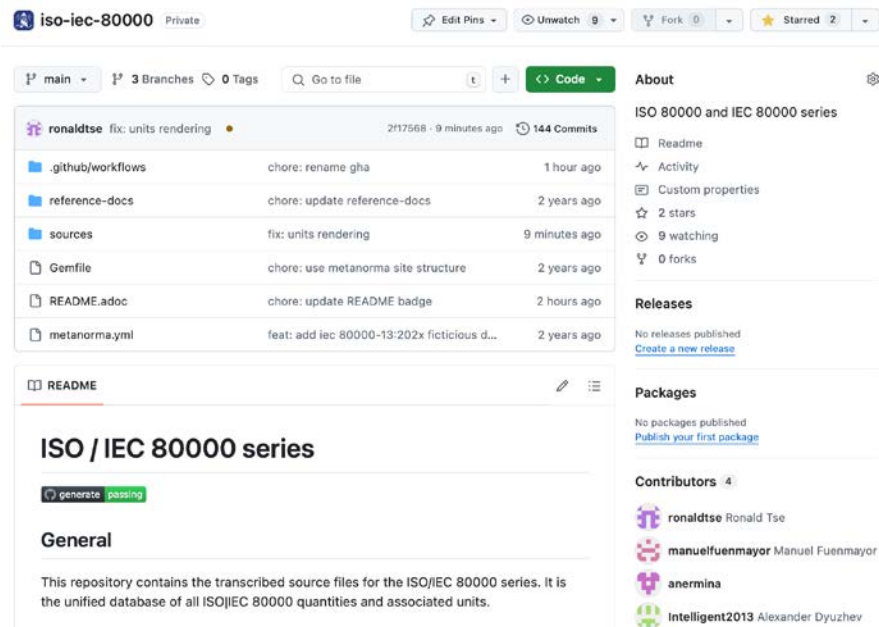
- TC VCARD
 - Common profile
 - Successor to vCard
 - VFORMAT
 - Format specification for iCalendar and vCard
 - VSCHEMA
 - Schema description language for iCalendar and vCard components
 - VPATH
 - Query and path specification for VFORMAT
- ISO 56100 Standardized profile
- Naming -- Structure of personal names (with ISO/TC 37)

Ongoing projects: Units

- TC UNITS
 - UnitsML: Machine-readable units specification
 - Revised UnitsDB
 - New liaison with ISO/TC 20/SC 6 “Standard atmosphere”
 - New projects on “Reference atmospheres”
 - ISO 2533 Standard atmosphere
 - ISO 5878 Temperature profiles, wind characteristics and humidity models
- {UnitsML presentation given separately}

ISO | IEC 80000 series in machine-readable form

- In ISO SMART, participated with the BIPM in developing a “SMART” version of the ISO | IEC 80000 series
- Project was completed and now to be presented to TC 12
- The resulting dataset is called the “ISQ dataset”



ISO | IEC 80000 Parts 2-13 in a unified dataset

- A unified dataset was created through manual encoding of all symbols, quantities and units of the ISO|IEC 80000 parts, into machine-readable data called the “ISO dataset”
- The ISQ dataset is then used to re-generate the documents as ISO|IEC 80000 parts, demonstrating that the TC only needs to manage machine-readable data, then the documents all follow automatically with no human input needed
- The ISQ dataset can be then used by information systems (e.g. linking with the Digital SI reference, UnitsML, etc)

Unified dataset

General

The unified dataset is a YAML file that contains all math symbols, quantities and units from the ISO/IEC 80000 series.

The unified dataset is a set of YAML files with a YAML schema defined in:

- `sources/dataset/schema-data.yaml`

The dataset consists of the following files:

`sources/dataset/math.yaml`

The unified dataset of math symbols used in ISO 80000-2. This file is used in the generation of ISO 80000-2.

`sources/dataset/quantities.yaml`

The unified dataset of all ISO/IEC 80000 quantities and associated units. This file is used for ISO/IEC 80000 Parts 3 to 13.

Note	Both dataset files conform to the same schema.
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The exception part is:

- Part 1, as it contains only general information and does not contain any quantities or units.

Note	The datasets are manually sourced from the ISO/IEC 80000 series.
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The ISQ dataset is used to generate a set of documents

ISO/IEC

GENERATED: 2025-05-22 METANORMA 2.1.9

ISO/IEC 80000 series

ISO 80000-1:2022 Edition: 2

Quantities and units – General

STAGE: 60.60 2022-12

[HTML](#) [PDF](#) [Word](#) [XML](#) [Relaton XML](#)

ISO 80000-10:2019 Edition: 2

Quantities and units – Atomic and nuclear physics

STAGE: 60.60 2019-08

[HTML](#) [PDF](#) [Word](#) [XML](#) [Relaton XML](#)

ISO 80000-11:2019 Edition: 2

Quantities and units – Characteristic numbers

STAGE: 60.60 2019-10

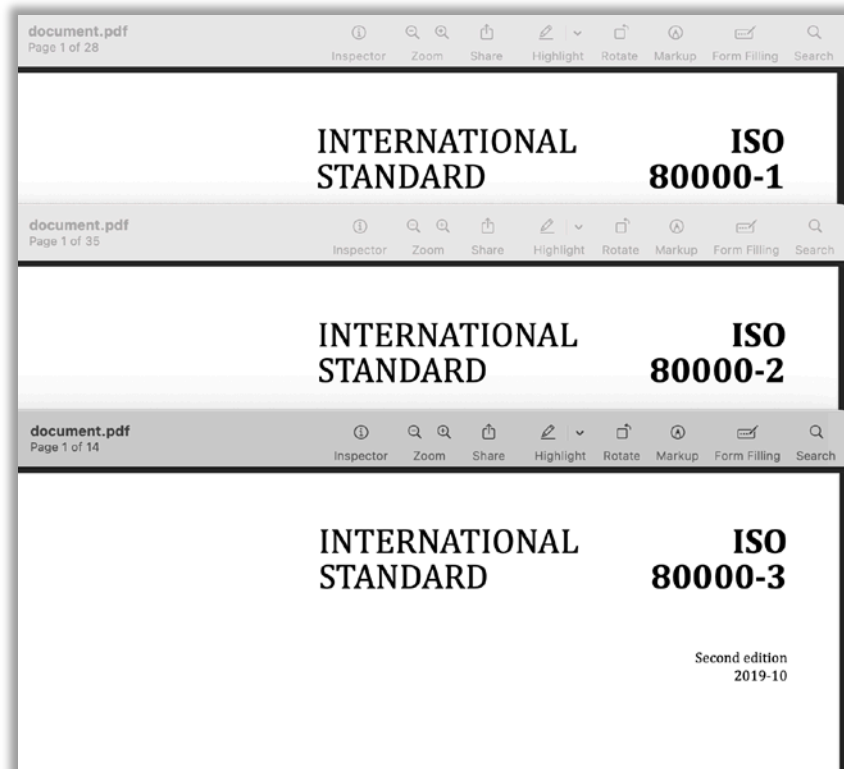
[HTML](#) [PDF](#) [Word](#) [XML](#) [Relaton XML](#)

ISO 80000-12:2019 Edition: 2

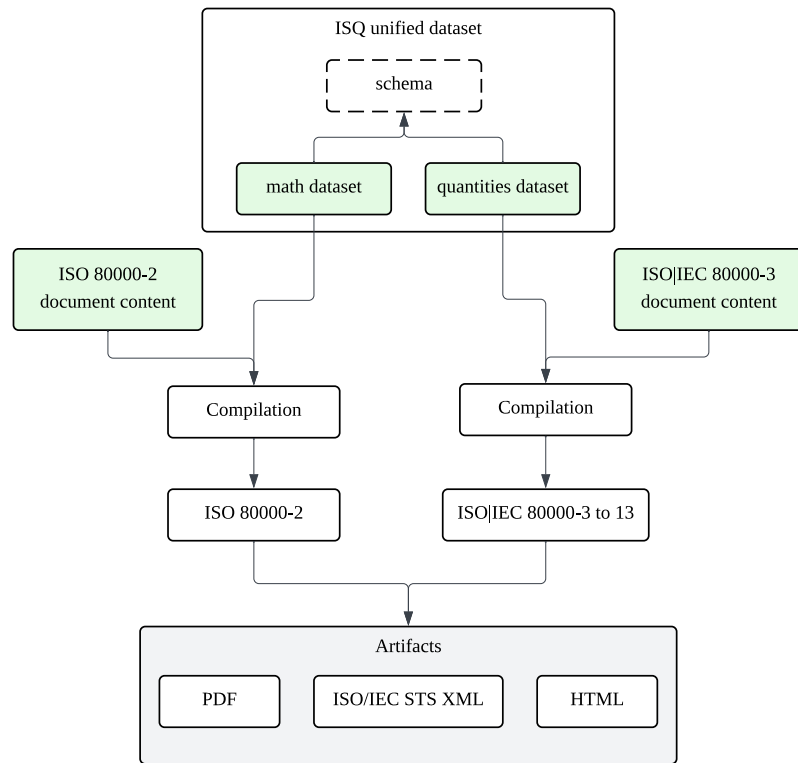
Quantities and units – Condensed matter physics

STAGE: 60.60 2019-08

[HTML](#) [PDF](#) [Word](#) [XML](#) [Relaton XML](#)



Workflow of generating standards from the unified dataset



Sample content generated from the unified dataset

Part 2 (symbols)

6 Sets

Table 2 — Symbols and expressions for sets

Item No.	Symbol expression	Meaning, verbal equivalent	Remarks and examples
2-6.1	$x \in A$	x belongs to A , x is an element of the set A	$A \ni x$ has the same meaning as $x \in A$.
2-6.2	$y \notin A$	y does not belong to A , y is not an element of the set A	$A \not\ni y$ has the same meaning as $y \notin A$. The negating stroke may also be vertical.
2-6.3	$\{x_1, x_2, \dots, x_n\}$	set with elements $x_1, x_2, \dots, x_n$	Also $\{x_i i \in I\}$, where I denotes a set of subscripts.
2-6.4	$\{x \in A p(x)\}$	set of those elements of A for which the proposition $p(x)$ is true	EXAMPLE $\{x \in \mathbf{R} x \geq 5\}$ (1) If it is clear from the context which set A is considered, the notation $\{x p(x)\}$ can be used (for example $\{x x \geq 5\}$, if it is clear that real numbers are considered). Instead of the vertical line often a colon is used as separator: $\{x \in A : p(x)\}$.
2-6.5	card A $ A $	number of elements in A , cardinality of A	The cardinality can be a transfinite number. The symbol $ $ is also used for absolute value of a real number (see Table 6), modulus of a complex number (see Table 11) and magnitude of a vector (see Table 14).

Part 3 (quantities + units)

Table 1 — Quantities and units of space and time

Item No.	Quantity			Unit	Remarks
	Name	Symbol	Definition		
3-1.1	length	l L	linear extent in space between any two points	m	Length does not need to be measured along a straight line. Length is one of the seven base quantities in the International System of Units (ISO 80000-1).
3-1.2	width breadth	b B	minimum length of a straight line segment between two parallel straight lines (in two dimensions) or planes (in three dimensions) that enclose a given geometrical shape	m	This quantity is non-negative.
3-1.3	height depth altitude	h H	minimum length of a straight line segment between a point and a reference line or reference surface	m	This quantity is usually signed. The sign expresses the position of the particular point with respect to the reference line or surface and is chosen by convention. The symbol H is often used to denote altitude, i.e. height above sea level.
3-1.4	thickness	d δ	width (Table 1)	m	This quantity is non-negative.
3-1.5	diameter	d D	width (Table 1) of a circle, cylinder or sphere	m	This quantity is non-negative.

Two parts of the dataset: math and quantities

Math symbols

The unified dataset of math symbols is sourced from the following parts of the ISO/IEC 80000 series:

- ISO 80000-2:2019 - Mathematical signs and symbols to be used in the natural sciences and technology

The tables are generated using the `table_generator.liquid` file located in the `sources/iso-80000-2-2019/sections` directory.

Note	For Clause 17 "Coordinate systems", the table has an extra column for "name", hence the Liquid template contains special handling for this case.
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Quantities

The unified dataset of all ISO/IEC 80000 quantities and associated units is sourced from the following parts of the ISO/IEC 80000 series:

- ISO 80000-3:2019 - Space and time
- ISO 80000-4:2019 - Mechanics
- ISO 80000-5:2019 - Thermodynamics
- IEC 80000-6:2022 - Electromagnetism (English and French versions)
- ISO 80000-7:2019 - Light
- ISO 80000-8:2020 - Acoustics
- ISO 80000-9:2019 (with Amendment 1) - Physical chemistry and molecular physics
- ISO 80000-10:2019 - Atomic and nuclear physics
- ISO 80000-11:2019 - Characteristic numbers
- ISO 80000-12:2019 - Solid state physics
- IEC 80000-13:202x - Information science and technology (English version)

The tables are generated using the `table_generator.liquid` file located in the `sources/dataset` directory.

Note	For Parts 6 and 13, both parts published by the IEC, contain bilingual content, and are also formatted slightly differently amongst the two. This is handled in the Liquid template.
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Both math and quantities data work with the same schema

Part 2 (symbols)

```
- part: 2-5
id: t2-5.1
num: 2-5.1
designations:
- designation:
  en:
    text: conjunction
    index_as:
      - conjunction
def:
  en: |
    conjunction of stem:[p] and stem:[q], stem:[p]
    and stem:[q]
symbols:
- p ^^ q
- part: 2-5
id: t2-5.2
num: 2-5.2
designations:
- designation:
  en:
    text: disjunction
    index_as:
      - disjunction
def:
  en: 'disjunction of stem:[p] and stem:[q], stem:[p] or stem:[q]'
symbols:
- p vv q
remarks:
  en: |
    This "or" is inclusive, i.e. stem:[p vv q] is
    true, if either stem:[p] or stem:[q], or both
    are true.
```

Part 13 (quantities + units)

```
- part: '13'
id: t13-1
num: 13-1
oldnum: "801"
designations:
- designation:
  en:
    text: traffic intensity
    index_as:
      - traffic intensity
  fr:
    text: intensité (f) de trafic
    index_as:
      - intensité de trafic
    noun: intensité
    gender: f
    adj: de trafic
symbols:
- ii(A)
def:
  en: number of simultaneously busy resources in a particular pool of
  fr: nombre de ressources simultanément occupées d'un ensemble donné
units:
- stem:["E"]
remarks:
  en: See <<iec60050-715,locality:item="715-05-02">>.
  fr: Voir la <<cei60050-715,droploc%locality:item="715-05-02">>.
```

Dataset content structure

- Glossarist is a concept description schema based on ISO 10241-1 (Terminology)
- A data schema that covers Part 2 (math) and Parts 3-13 (quantities) is created based on the Glossarist schema
- Every entry of the “math” and “quantities” dataset follow the YAML schema described at the right (partial)

```
1  # yaml-language-server: $schema=https://json-schema.org/draft/2020-12/schema
2  ---
3  $schema: "http://json-schema.org/draft-04/schema#"
4  title: ISO/IEC 80000 quantities schema
5  description: "YAML Schema for ISO/IEC 80000 quantities YAML files."
6  $defs:
7    designation:
8      description: "Designation of the quantity (multilingual)."
```

```
9      type: object
10     additionalProperties: false
11     properties:
12       designation:
13         type: object
14         properties:
15           en:
16             $ref: "#/$defs/designation_lang"
17           fr:
18             $ref: "#/$defs/designation_lang"
19     designation_lang:
20       description: "Designation in a specific language."
```

```
21       type: object
22       required:
23         - text
24       additionalProperties: false
25       properties:
26         text:
27           type: string
28         index_as:
```

Thank you, questions welcome!